

Gabriele Berton

 [Google Scholar](#)

 [GitHub](#)

 [Personal Website](#)

Research Engineer at Google DeepMind

Training VLMs at Google DeepMind. PhD with 8 first-author CVPR/ICCV/ECCV papers. 3000+ GitHub stars.

EDUCATION & EMPLOYMENT

Research Engineer | **Google DeepMind** | Mar 2026 – present

- Working on foundational Vision-Language Models, at the intersection between research and development

Postdoctoral scientist | **Amazon** | Mar 2025 – Mar 2026

- Worked on VLMs and LLMs

Visiting PhD Student | **Carnegie Mellon University** | Mar 2024 – Jul 2024

- Worked on 3D reconstruction

PhD | Polytechnic of Turin | Feb 2021 – Nov 2024

- Created AstroLoc (ICCV 2025), a model for Astronaut Photography Localization, **currently used by NASA**. AstroLoc provides a **10x speed up** and **increases accuracy by 21%** to NASA's previous model.
- First author of CosPlace (CVPR 2022), a Visual Place Recognition model that achieved SOTA with **8x smaller descriptor**, **75% less VRAM**, same training time and 1000x more scalability than previous SOTA.
- Developed a large-scale (**over one billion images**) retrieval software for Visual Place Recognition
- Co-supervised 12 master students
- Teaching assistant of “Machine Learning and Deep Learning” and “Algorithms and Programming”

Applied Scientist Intern | **Amazon Berlin** | May 2022 – Sep 2022

- Worked on an action spotting pipeline in videos through multi-modal neural networks
- Analyzed video understanding and audio processing methods for a real-world application

Applied Scientist | Italian Institute of Technology & Polytechnic of Turin | Apr 2020 – Feb 2021

Software Engineer | Consoft Sistemi | Dec 2017 – Dec 2018

INVITED TALKS

- Jan 2024 - Talk on Applications and Challenges of Visual Geo-localization, Visual Recognition Group at the Czech Technical University, Czechia
- Jul 2023 - Talk on Scalable Visual Geo-localization, Pattern Analysis and Computer Vision group at the Italian Institute of Technology, Italy
- Jun 2023 - Talk on Recent Image Geo-localization Benchmark and Large-scale Real-world Scenarios, **CVPR Tutorial** on A Comprehensive Tour and Recent Advancements toward Real-world Visual Geo-Localization, Canada
- Jun 2023 - Machine Learning Genoa Center, University of Genoa, Italy

AWARDS

- Oct 2025 - 2nd prize at the CroCoDL challenge at ICCV
- Sep 2025 - Outstanding reviewer at ICCV
- May 2024 - Awarded with Blanceflor Foundation Fellowship (30.000 SEK)
- Aug 2023 - Awarded with ESA NoR sponsorship (3.000 €)

- Jul 2023 - Best Student Research Award in the Video&Poster Competition at EuroCon 2023 (500 €)

SELECTED PUBLICATIONS (FIRST AUTHOR AT TOP CONFERENCES)

- **ICCV 2025** - AstroLoc: Robust Space to Ground Image Localizer
G. Berton, A. Stoken, C. Masone
- **ECCV 2024** - MeshVPR: Citywide Visual Place Recognition Using 3D Meshes
G. Berton, L. Junglas, R. Zacccone, T. Pollock, B. Caputo, C. Masone
- **CVPR 2024** - EarthLoc: Astronaut Photography Localization by Indexing Earth from Space
G. Berton, A. Stoken, B. Caputo, C. Masone
- **RAL/ICRA 2024** - JIST: Joint Image and Sequence Training for Sequential Visual Place Recognition
G. Berton*, G. Trivigno*, B. Caputo, C. Masone
- **ICCV 2023** - EigenPlaces: Training Viewpoint Robust Models for Visual Place Recognition
G. Berton*, G. Trivigno*, C. Masone, J. Aragon, B. Caputo
- **ICCV 2023** - Divide&Classify: Fine-Grained Classification for City-Wide Visual Geo-localization
G. Trivigno*, G. Berton*, C. Masone, B. Caputo
- **CVPR 2022 (Oral)** - Deep Visual Geo-localization Benchmark
G. Berton, R. Mereu, G. Trivigno, C. Masone, G. Csurka, T. Sattler, B. Caputo
- **CVPR 2022** - Rethinking Visual Geo-localization for Large-Scale Applications
G. Berton, C. Masone, B. Caputo
- **ICCV 2021** - Viewpoint Invariant Dense Matching for Visual Geolocalization
G. Berton, C. Masone, V. Paolicelli, B. Caputo

OTHER EXPERIENCES

- Ex-member of RoboTo, my university's robotics competition team, integrating computer vision into robots.
- Finalist of the Italian Math Olympiad, individually and with my high-school team.
- Been to 50+ countries: a year in China; worked six months on Australian farms, six months from Thailand to Italy without flying.
- Recruited 60 international English teachers for a summer camp in Belarus.